

Maintenance Operations

Turning Manual Maintenance Reporting Into a Clear Operational View

Project Overview

Role: Business Analyst + Data Analyst

Period: August to November 2024

Primary Stakeholder: Maintenance Manager

Data: Approximately 10,000 historical records across 20 teams

Solution: Power BI dashboard built from scratch

Tools: Power BI, Excel, SQL, Python, Power Query, Google Sheets and DAX

The Problem

The Maintenance Manager had a reporting problem that looked simple at first.

There was plenty of maintenance data, but getting a clear picture of what was actually happening required too much manual work.

Information was spread across Excel files, CSV files, Google Sheets and database sources. Every reporting cycle involved bringing updates together before questions such as these could be answered:

How much work has actually been completed?

What is overdue?

Which teams have the largest outstanding workload?

Are high-priority jobs still open?

Are certain maintenance problems happening repeatedly?

The data existed.

The problem was turning it into something the manager could use without rebuilding the same report every week.

That became the focus of the project.

My Role

I worked across both the business and data sides of the project.

On the BA side, I helped translate the Maintenance Manager's reporting needs into actual measures, business rules and reporting requirements.

On the data side, I worked with the historical records, cleaned and structured the information, analyzed maintenance performance and built the Power BI dashboard.

My work covered:

- understanding the reporting requirement;
- defining KPIs and business rules;
- cleaning and structuring the data;
- analyzing maintenance performance;
- identifying exceptions and operational patterns;
- building the Power BI solution from scratch;
- validating the calculations and filters;
- documenting the solution;
- presenting it to the stakeholder;
- providing a walkthrough and training.

The project wasn't just about building charts. The important part was deciding **what the manager needed to see and how each measure should be defined.**

Understanding the Data

The dataset contained approximately **10,000 historical records across 20 teams.**

The records covered:

- maintenance tasks;
- work orders;
- inspections;
- service requests;
- equipment records.

Some of the useful fields included:

Work Order / Task ID

Team

Technician

Asset / Equipment

Created Date

Due Date

Completion Date

Status

Priority

Category

Location

Remarks

Before I could build meaningful KPIs from these fields, the data itself needed attention.

The First Problem Was Data Quality

The source data wasn't perfectly consistent.

I found issues such as:

- duplicates;
- missing values;
- inconsistent team names;
- inconsistent date formats;
- different status naming;
- blank records.

For example, if one source contains:

Team A

TEAM A

Team-A

Team A

those values may represent the same team but appear separately in a report unless they are standardized.

The same problem applies to statuses.

Completed

Complete

COMPLETED

If these are treated as different categories, even a visually attractive dashboard can produce misleading results.

So before building the reporting layer, I cleaned and structured the data so that the analysis was based on consistent information. The underlying report confirms these were among the actual quality issues handled during the engagement.

Moving From Data to Business Questions

Once the data was usable, I didn't start by asking:

"Which Power BI visuals should I create?"

I started with the questions the Maintenance Manager needed answered.

I organized the analysis around:

Completion

How much assigned maintenance work has been completed?

Overdue Work

Which jobs have passed their due date but are still not completed?

Work Status

How much work is completed, open or pending?

Priority

Are important maintenance jobs still outstanding?

Team Performance

How does maintenance activity vary between teams?

Technician Performance

How is work distributed and completed across technicians?

Backlog

What work remains outstanding?

Workload

Is work distributed evenly across the operation?

Categories

Are particular types of maintenance appearing repeatedly?

Equipment

Are certain assets or equipment showing recurring patterns?

This gave the dashboard a purpose before I started designing it.

Defining the Business Rules

One of the most important parts of the project was agreeing on what the numbers actually meant.

For example, the completion rate was defined as:

Completed Maintenance Tasks / Total Assigned Maintenance Tasks × 100

The analysis showed an overall completion rate of:

92%

That does **not** mean I improved maintenance completion by 92%.

It means that **92% was the completion rate observed in the analyzed data.**

Keeping that distinction clear matters when presenting analytical work.

Defining Overdue Work

"Overdue" also needed a clear definition.

The rule used was:

Due Date < Current Date

AND

Status != Completed

This sounds straightforward, but defining it properly prevents interpretation differences.

Without a rule, one person might count an incomplete task due today as overdue while another might not.

Once the definition is agree

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